

Loan Defaulter Analysis

Background:

In this case study, we will analyze a dataset from a financial institution that deals with loans. The dataset includes various details about the loan applicants and their financial status. The trainees, with a basic understanding of Excel and data visualization tools, will explore this dataset to uncover insights and make informed decisions.

Problem Statement

What is the problem?

The financial institution wants to understand the factors affecting loan status and identify patterns that might help in assessing loan applications more effectively. The goal is to minimize risk and improve loan approval processes.

Why is it important to solve it?

Efficient loan processing and decision-making are vital for financial institutions to minimize risk and maximize profit. Understanding the key factors that influence loan approval and defaults can help in creating better strategies for assessing loan applications.

Data Link

<https://www.kaggle.com/datasets/manishtrpathi86/deloitte-hackathon-predict-the-loan-defaulter?select=train.csv>

Data Dictionary

- ID: unique ID of representative
- Loan Amount: loan amount applied
- Funded Amount: loan amount funded
- Funded Amount Investor: loan amount approved by the investors
- Term: term of loan (in months)
- Batch Enrolled: batch numbers to representatives
- Interest Rate: interest rate (%) on loan
- Grade: grade by the bank
- Sub Grade: sub-grade by the bank
- Employment Duration: duration
- Home Ownership: Owner ship of home
- Verification Status: Income verification by the bank
- Payment Plan: if any payment plan has started against loan
- Loan Title: loan title provided
- Debit to Income: ratio of representative's total monthly debt repayment divided by self reported monthly income excluding mortgage
- Delinquency - two years: number of 30+ days delinquency in past 2 years
- Inquires - six months: total number of inquiries in last 6 months
- Open Account: number of open credit line in representative's credit line

- Public Record: number of derogatory public records
- Revolving Balance: total credit revolving balance
- Revolving Utilities: amount of credit a representative is using relative to revolving_balance
- Total Accounts: total number of credit lines available in representatives credit line
- Initial List Status: unique listing status of the loan - W(Waiting), F(Forwarded)
- Total Received Interest: total interest received till date
- Total Received Late Fee: total late fee received till date
- Recoveries: post charge off gross recovery
- Collection Recovery Fee: post charge off collection fee
- Collection 12 months Medical: total collections in last 12 months excluding medical collections
- Application Type: indicates when the representative is an individual or joint
- Last week Pay: indicates how long (in weeks) a representative has paid EMI after batch enrolled
- Accounts Delinquent: number of accounts on which the representative is delinquent
- Total Collection Amount: total collection amount ever owed
- Total Current Balance: total current balance from all accounts
- Total Revolving Credit Limit: total revolving credit limit
- Loan Status: 1 = Defaulter, 0 = Non Defaulters

Basic-Level Questions

1. How many loans are there in the dataset?
Hint: Use a simple count function.
2. What is the average loan amount?
Hint: Calculate the mean of the Loan Amount column.
3. What are the most common employment durations among applicants?
Hint: Use value counts on Employment Duration.
4. How many unique grades and subgrades are there in the dataset?
Hint: Count unique values in Grade and Sub Grade.
5. What is the distribution of loan terms (in months)?
Hint: Create a frequency table for the Term column.
6. What is the average interest rate for each loan grade?
Hint: Group by Grade and calculate the mean of Interest Rate.
7. How does the funded amount compare to the loan amount on average?
Hint: Compare the averages of Loan Amount and Funded Amount.
8. What are the top 5 loan purposes in the dataset?
Hint: Use value counts on the relevant column for loan purpose.
9. What proportion of loans are fully funded (i.e., Funded Amount equals Loan Amount)?
Hint: Calculate the ratio of loans where Funded Amount equals Loan Amount.
10. What is the distribution of loans across different loan statuses?
Hint: Use value counts on Loan Status.

Medium-Level Questions

11. What is the correlation between loan amount and interest rate?
Hint: Compute the correlation coefficient.
12. How does the average total current balance vary across different employment durations?
Hint: Group by Employment Duration and calculate mean of Total Current Balance.
13. Is there a significant difference in funded amount investor between different loan grades?
Hint: Use group by and statistical tests.
14. What is the average last week pay across different loan statuses?
Hint: Group by Loan Status and calculate the mean of Last week Pay.
15. How does the loan amount vary across different loan purposes?
Hint: Group by loan purpose and analyze the loan amount.
16. Analyze the distribution of Interest Rate for different terms of loans.
Hint: Use boxplots or other visualization techniques.
17. What are the top 5 batches enrolled in terms of loan count?
Hint: Count loans per Batch Enrolled.
18. How does the total revolving credit limit vary with loan grades?
Hint: Group by Grade and analyze Total Revolving Credit Limit.
19. Is there a trend in the number of accounts delinquent over different loan grades?
Hint: Analyze Accounts Delinquent with respect to Grade.
20. What factors are most strongly correlated with the funded amount investor?
Hint: Use correlation matrix to identify strong correlations.

Advanced-Level Questions

21. Can you identify patterns in loan status based on the combination of term and grade?
Hint: Create a pivot table or multi-dimensional group by.
22. Analyze if there is any anomaly in interest rates for different employment durations.
Hint: Look for outliers or unusual patterns in the data.
23. How does the ratio of funded amount to loan amount vary across different loan statuses?
Hint: Calculate the ratio and analyze it by loan status.
24. Investigate the relationship between loan amount and total collection amount.
Hint: Consider scatter plots or correlation analysis.

Each of these questions is designed to progressively build the trainees' analytical skills, starting from basic data understanding to more complex analysis and insight generation.

Deliverables

- Case Study Document: Includes problem statement, data dictionary, and questions.
- Solution Guide: Detailed answers and explanations for each question.
- Additional Resources: References for further exploration.

Desired Outcome

The trainees will develop an analytical and logical mindset, understanding the importance of various factors in loan analysis. They will learn to apply different data analysis techniques to uncover insights and make data-driven decisions.